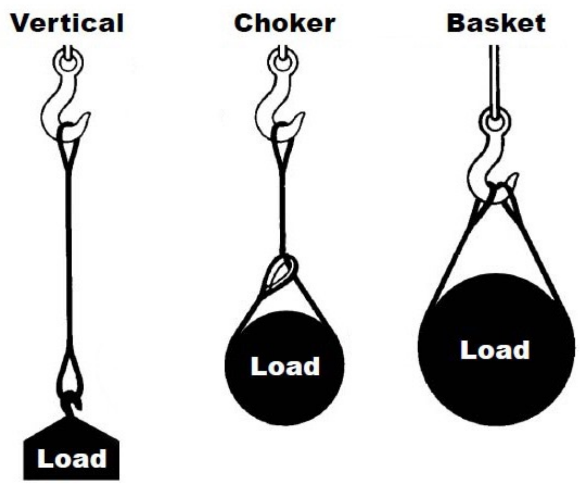
WIRE ROPE SLINGS & ASSEMBLIES

Mechanically Spliced Slings
6 & 8 Part Braided Slings
Cable Laid Slings
Hand Spliced Slings
Hand & Mechanical Spliced Grommets
Swage Socket Assemblies
Spelter Socket Assemblies

CoreLifting.com

General Information
Basic Hitches



VERTICAL, or straight, attachment is simply using a sling to connect a lifting hook to a load. Full rated lifting capacity of the sling may be utilized but must not be exceeded. A tagline should be used to prevent load rotation, which may damage a sling. When two or more slings are attached to the same lifting hook, the total hitch becomes, in effect, a lifting bridle, and the load is distributed equally among the individual slings.

CHOKER hitches reduce lifting capability of a sling since this method of rigging affects ability of the wire rope components to adjust during the lift. A choker is used when the load will not be seriously damaged by the sling body — or the sling damaged by the load, and when the lift requires the sling to snug up against the load.

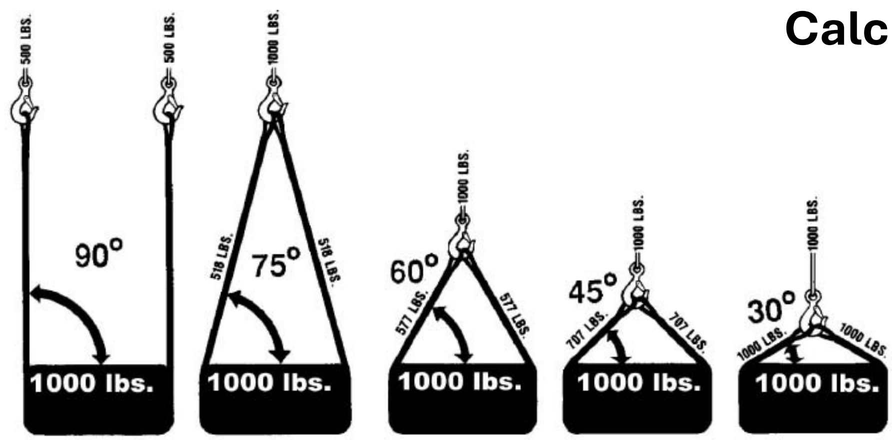
The diameter of the bend where the sling contacts the load should keep the point of choke against the sling BODY — never against a splice or the base of the eye. When a choke is used at an angle of less than 120 degrees (see next page), the sling-rated capacity must be adjusted downward.

A choker hitch should be pulled tight before a lift is made — NOT PULLED DOWN DURING THE LIFT. It is also dangerous to use only one choker hitch to lift a load which might shift or slide out of the choke.

BASKET hitches distribute a load equally between the two legs of a sling ... within limitations described above.

General Information
Load Calculating

Calculating the Load on Each Leg of a Sling



As the included angle between the legs of a sling decreases, the load on each leg increases. The effect is the same whether a single sling is used as a basket or two slings are used with each in a straight pull, as with a 2-legged bridle.

Anytime pull is exerted at an angle on a leg—or legs—of a sling, the load per leg can be determined by using the data in the table above. Proceed as follows to calculate this load—and determine the rated capacity required of the sling, or slings, needed for a lift.

1.First, divide the total load to be lifted by the number of legs to be used. This provides the load per leg if the lift were being made with all legs being vertically.

2.Determine the angle between the legs of the sling and the horizontal.

3.Then MULTIPLY the load per leg (as computed above) by the Load Factor for the leg angle being used (from the table at the bottom) – to compute the ACTUAL LOAD on each leg for this lift and angle. THE ACTUAL LOAD MUST NOT EXCEED THE RATED SLING CAPACITY.

Thus, in the above drawing (sling angle at 60°): 1000 ÷ 2 = 500 (Load Per Leg if a vertical lift) 500 x 1.154 = 577 lbs. – ACTUAL LOAD on each leg at the 60° included angle being used.

In the above drawing (sling angle of 45°): 1000 ÷ 2 = 500 (Load Per Leg if a vertical lift) 500 x 1.414 = 707 lbs. = ACTUAL LOAD on each leg at the 45° horizontal angle being used.

General Information
D/d Ratio

D/d Ratios Apply to Slings

When rigged as a basket, DIAMETER of the bend where a sling contacts the load can be a limiting factor on sling capacity. Standard D/d ratios—where "D" is the diameter of bend, and "d" the diameter of the rope—are applied to determine efficiency of various sling constructions, as indicated below:

•**Mechanically Spliced, Single-Part Slings:** 25 times rope diameter

•**Hand Spliced, Single-Part Slings:** 15 times rope diameter.

•**Braided Multi-Part Slings of 6 Parts:** 25 times component rope diameter.

•**Braided Multi-Part Slings of 8 Parts:** 25 times component rope diameter.

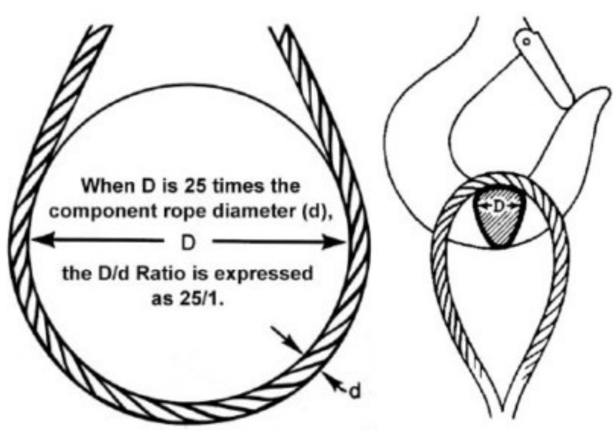
•**Cable Laid Slings:** 10 times sling body diameter.

•**Hand Tucked Grommets & Mechanically Joined Grommets:** 5 times sling body diameter.

Sling Eye Design

Sling eyes are designed to provide what amount to "small inverted slings" at the ends of the sling body. Therefore, the width of the eye opening will be affected by the same general forces which apply to legs of a sling rigged as a basket.

A sling eye should never be used over a hook or pin with a body diameter larger than the natural width of the eye. Never force an eye onto a hook. On the other hand, the eye should always be used on a hook or pin with at least the nominal diameter of the rope—since applying the D/d Ratio shows an efficiency loss of approximately 50% when the relationship is less than 1/1.



Sling Body Construction
Determined by Application

Sling Usage Dictates Sling Body Construction

Whether to use a single-part sling (one made of a single wire rope in the sling body) or a multi-part sling (several ropes in the body) is usually the first decision to make after determining the sling length and capacity for a lift. The starting point for this decision involves the handling characteristics of the sling more than any other factor. Based on capacity alone, multi-part slings will be more flexible…more easily handled…than single-part slings. The larger the capacity of a sling, the more important this becomes…to the point, it becomes unrealistic to build large capacity slings from single, very large wire ropes.

Multi-part slings provide the only practical means for obtaining extremely heavy lift capacity … in hundreds of tons. Various approaches to building multi-part slings have been developed; the most common of these are Braided or Multi-Part.

Braided, or multi-part, slings are often selected because this construction provides a gripping effect, which reduces load slippage and rotation.
In the design of the sling, rope engineers must seek a balance between strength-handling characteristics and number of parts…since there is a tendency to lose strength as core parts are added to increase flexibility.

Design Considerations
Construction & Length

Sling Usage Dictates Sling Body Construction

Whether to use a single-part sling (one made of a single wire rope in the sling body) or a multi-part sling (several ropes in the body) is usually the first decision to make after determining the sling length and capacity for a lift. The starting point for this decision involves the handling characteristics of the sling more than any other factor. Based on capacity alone, multi-part slings will be more flexible...more easily handled...than single-part slings. The larger the capacity of a sling, the more important this becomes...to the point, it becomes unrealistic to build large capacity slings from single, very large wire ropes.

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Minimum Sling Body Length

This is the length of wire rope between splices, sleeves or fittings. Generally, the minimum body length is equal to ten (10) times the sling body diameter. This allows approximately one and one half (1-1/2) rope lays between splices. For Multi-part slings, the minimum body length between splices is equal to forty (40) times the component rope diameter.

General Information

Bridle Legs – Angle Effects

Angles of Bridles

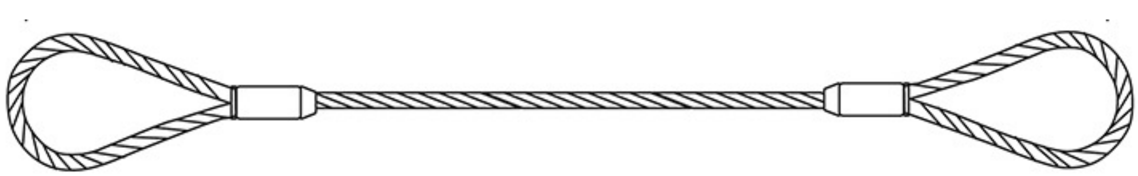
The horizontal angle of bridles with 3 or more legs is measured the same as the horizontal sling angle of 2-legged hitches. In this case, where a bridle designed with different leg lengths results in horizontal angles, the leg with the smallest horizontal angle will carry the greatest load. Therefore, the smallest horizontal angle is used in calculating actual leg load and evaluating the rated capacity of the sling proposed.

Leg Angle (Degrees)	Load Factor
90	1.000
85	1.003
80	1.015
75	1.035
70	1.064
65	1.103
60	1.154
55	1.220
50	1.305
45	1.414
40	1.555
35	1.743
30	2.000

Mechanically Spliced
1-Leg Slings

Construction	Rope Diameter (inches)	Rated Capacity—Tons *						Minimum Sling Length (When Using Standard Eyes)	Standard Eye Size (inches) W x L	Thimble Eye Size (inches) W x L	Eye Hook Cap. Tons
		Vertical	** Choker Hitch	Basket Hitch							
				Vertical (U)	60°	45°	30°				
6 x 19 IWRC	1/4	.65	.48	1.3	1.1	.91	.65	1' 6"	2 x 4	7/8 x 1-5/8	3/4
6 x 19 IWRC	5/16	1	.74	2	1.7	1.4	1	1' 9"	2-1/2 x 5	1-1/16 x 1-7/8	1
6 x 19 IWRC	3/8	1.4	1.1	2.9	2.5	2	1.4	2'	3 x 6	1-1/8 x 2-1/8	1-1/2
6 x 19 IWRC	7/16	1.9	1.4	3.9	3.4	2.7	1.9	2' 3"	3-1/2 x 7	1-1/4 x 2-1/4	2
6 x 19 IWRC	1/2	2.5	1.9	5.1	4.4	3.6	2.5	2' 6"	4 x 8	1-1/2 x 2-3/4	3
6 x 19 IWRC	9/16	3.2	2.4	6.4	5.5	4.5	3.2	2' 9"	4-1/2 x 9	1-1/2 x 2-3/4	3
6 x 19 IWRC	5/8	3.9	2.9	7.8	6.8	5.5	3.9	3'	5 x 10	1-3/4 x 3-1/4	5
6 x 19 IWRC	3/4	5.6	4.1	11	9.7	7.9	5.6	3' 6"	6 x 12	2 x 3-3/4	5
6 x 19 IWRC	7/8	7.6	5.6	15	13	11	7.6	4'	7 x 14	2-1/4 x 4-1/4	7-1/2
6 x 19 IWRC	1	9.8	7.2	20	17	14	9.8	4' 6"	8 x 16	2-1/2 x 4-1/2	10
6 x 19 IWRC	1-1/8	12	9.1	24	21	17	12	5'	9 x 18	2-7/8 x 5-1/8	10
6 x 36 IWRC	1-1/4	15	11	30	26	21	15	5' 6"	10 x 20	2-7/8 x 5-1/8	15
6 x 36 IWRC	1-3/8	18	13	36	31	25	18	6'	11 x 22	3-1/2 x 6-1/4	15
6 x 36 IWRC	1-1/2	21	16	42	37	30	21	7'	12 x 24	3-1/2 x 6-1/4	AH-22
6 x 36 IWRC	1-5/8	24	18	48	42	35	24	8'	13 x 26	3-1/2 x 6-1/4	AH-30
6 x 36 IWRC	1-3/4	28	21	57	49	40	28	8'	14 x 28	4 x 8	AH-37
6 x 36 IWRC	2	37	28	73	63	52	37	9'	16 x 32	4-1/2 x 9	AH-45
6 x 36 IWRC	2-1/4	44	35	89	77	63	44	10'	18 x 36	6 x 12	AH-45
6 x 36 IWRC	2-1/2	54	42	109	94	77	54	11'	20 x 40	7 x 14	AH-60
6 x 36 IWRC	2-3/4	65	51	130	113	92	65	12'	22 x 44	–	–
6 x 36 IWRC	3	77	60	153	133	108	77	13'	24 x 48	–	–
6 x 36 IWRC	3-1/2	102	79	203	143	144	102	14'	28 x 56	–	–

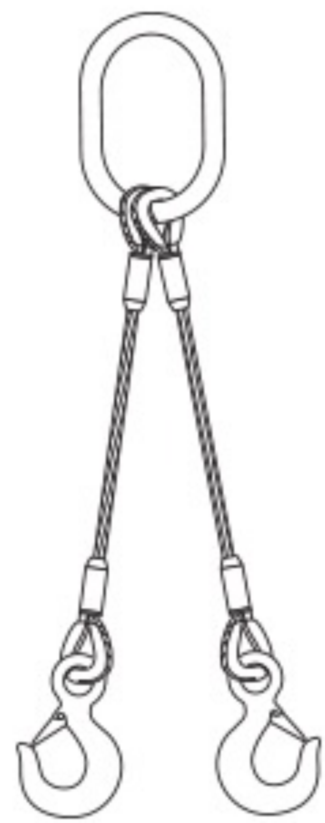
- *Rated capacity in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 25.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Rated capacities based on EIPS wire rope.
- Sling angles less than 30° shall not be used.
- **See choker hitch rated capacity adjustment.



Mechanically Spliced
2-Leg Slings

Rope Diameter 6 x 19 IWRC 6 x 36 IWRC	Min. Sling Length When Using Std. Eyes	Rated Capacity Tons * 2-Leg Choker Hitch ***			Alloy Oblong Link			Hook		
		60°	45°	30°	D	L	W	WLL **	E	R
1/4	1' 3"	1.1	.91	.65	1/2	5	2-1/2	3/4	.90	4.42
5/16	1' 6"	1.7	1.4	1	5/8	6	3	1	.90	4.42
3/8	1' 8"	2.5	2	1.4	3/4	5-1/2	2-3/4	1-1/2	.93	5.64
7/16	1' 10"	3.4	2.7	19.	7/8	6	3	2	1	5.64
1/2	2' 0"	4.4	3.6	2.5	3/4	5-1/2	2-3/4	3	1.13	6.39
9/16	2' 2"	5.5	4.5	3.2	1	8	4	3	1.13	6.39
5/8	2' 4"	6.8	5.5	3.9	1	8	4	5	1.47	7.90
3/4	2' 9"	9.7	7.9	5.6	1	8	4	5	1.47	7.90
7/8	3' 3"	13	11	7.6	1-1/4	8-3/4	4-3/8	7	1.75	10.09
1	3' 6"	17	14	9.8	1-1/4	8-3/4	4-3/8	11	2.29	12.43
1-1/8	4' 0"	21	17	12	1-1/2	12	6	11	2.29	12.43
1-1/4	4' 6"	26	21	15	1-3/4	12	6	15	2.50	13.94
1-3/8	5' 0"	31	25	18	1-3/4	12	6	15	2.50	13.94
1-1/2	5' 6"	37	30	21	2	14	7	22	3.30	17.09
1-3/4	9'	49	40	28	2-1/4	8	16	37	4.25	24.81
2	10'	56	52	37	2-1/2	8	16	45	4.75	27.44

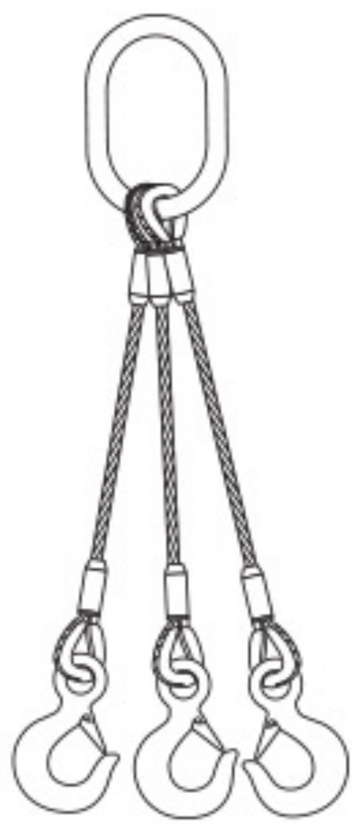
- *Rated capacity in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 25.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Rated capacities based on EIPS wire rope.
- Sling angles less than 30° shall not be used.
- Rated capacity for two-legged wire rope sling bridles, whether used as chokers or hook or other end fixtures, is affected by rigging angles, the same as straight slings in basket hitches.
- Note: Reduction in rated capacity as legs spread to wider lifting configurations.
- **Working load limit in tons – 2,000 lbs per ton.
- *** See choker hitch rated capacity adjustment.



Mechanically Spliced
3-Leg Slings

Rope Diameter 6 x 19 IWRC 6 x 36 IWRC	Min. Sling Length When Using Std. Eyes	Rated Capacity Tons *			Alloy Oblong Link			Hook			Carbon Pear Link			
		60°	45°	30°	D	L	W	WLL **	E	R	D	A	B	C
1/4	1' 3"	1.7	1.4	.97	1/2	5	2-1/2	3/4	15/16	4.42	1/2	5	2-1/2	2-1/2
5/16	1' 6"	2.6	2.1	1.5	1/2	5	2-1/2	1	1-1/32	4.42	5/8	6	3	3
3/8	1' 8"	3.7	3	2.2	5/8	6	3	1-1/2	1-1/16	5.64	3/4	5-1/2	2-3/4	2-3/4
7/16	1' 10"	5	4.1	2.9	3/4	5.5	2.75	2	1-7/32	5.64	7/8	6	3	3
1/2	2' 0"	6.6	5.4	3.8	7/8	5.5	2.75	3	1-1/2	6.39	3/4	5-1/2	2-3/4	2-3/4
9/16	2' 2"	8.3	6.8	4.8	1	7	3.5	3	1-1/2	6.39	1	8	4	4
5/8	2' 4"	10	8.3	5.9	1	7	3.5	5	1-7/8	7.90	1	8	4	4
3/4	2' 9"	15	12	8.4	1-1/4	8.75	4.38	5	1-7/8	7.90	–	–	–	–
7/8	3' 3"	20	16	11	1-1/2	10.5	5-1/2	7-1/2	2-1/4	10.09	–	–	–	–
1	3' 6"	26	21	15	1-1/2	10.5	5-1/2	11	2-1/2	12.43	–	–	–	–
1-1/8	4' 0"	31	26	18	1-3/4	12	6	11	2-1/2	12.43	–	–	–	–
1-1/4	4' 6"	38	31	22	2	14	7	15	3-3/8	13.94	–	–	–	–
1-3/8	5' 0"	46	38	27	2-1/4	16	8	15	3-3/8	13.94	–	–	–	–
1-1/2	5' 6"	55	45	32	2-1/4	16	8	AH-22	3-3/8	13.94	–	–	–	–
1-5/8	8'	63	52	37	3-1/4	10	20	AH-30	4	14-1/6	–	–	–	–
1-3/4	9'	74	60	42	3-1/4	10	20	AH-37	4-1/4	18-5/16	–	–	–	–

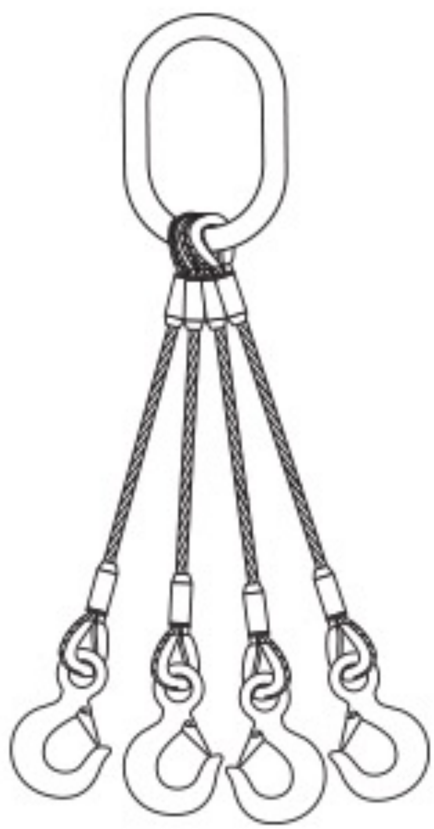
- *Rated capacity in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 25.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Rated capacities based on EIPS wire rope.
- Sling angles less than 30° shall not be used.
- Sling angles in this table depart from the traditional method of vertical angles measured at the sling hook. It has long been the opinion of sling users that it is easier to measure a sling angle relative to the ground or horizontal. The method is the same whichever angle is used. When the horizontal angle is used, you must use the trigonometric cosine of the vertical angle.
- **Working load limit in tons – 2,000 lbs per ton.



Mechanically Spliced
4-Leg Slings

Rope Diameter 6 x 19 IWRC 6 x 36 IWRC	Min. Sling Length When Using Std. Eyes	Rated Capacity Tons *			Alloy Oblong Link			Hook		
		60°	45°	30°	D	L	W	WLL **	E	R
1/4	1' 3"	2.2	1.8	1.3	1/2	5	2-1/2	3/4	.90	4.42
5/16	1' 6"	3.5	2.8	2	5/8	6	3	1	.90	4.42
3/8	1' 8"	5	4.1	2.9	7/8	7.5	3.75	1-1/2	.93	5.64
7/16	1' 10"	6.7	5.5	3.9	7/8	7.5	3.75	2	1	5.64
1/2	2' 0"	8.8	7.1	5.1	1	7	3.56	3	1.13	6.39
9/16	2' 2"	11	9	6.4	1-1/4	8-3/4	4.38	3	1.13	6.39
5/8	2' 4"	14	11	7.8	1-1/4	8-3/4	4.38	5	1.47	7.90
3/4	2' 9"	19	16	11	1-1/2	11-1/2	5.25	5	1.47	7.90
7/8	3' 3"	26	21	15	1-3/4	12	6	7	1.75	10.09
1	3' 6"	34	28	20	2	14	7	11	2.29	12.43
1-1/8	4' 0"	42	34	24	2-1/2	16	8	11	2.29	12.43
1-1/4	4' 6"	51	42	30	3	18	9	15	2.50	13.94

- *Rated capacity in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 25.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Rated capacities based on EIPS wire rope.
- Sling angles less than 30° shall not be used.
- Sling angles in this table depart from the traditional method of vertical angles measured at the sling hook. It has long been the opinion of sling users that it is easier to measure a sling angle relative to the ground or horizontal. The method is the same whichever angle is used. When the horizontal angle is used, you must use the trigonometric cosine of the vertical angle.
- **Working load limit in tons – 2,000 lbs per ton.



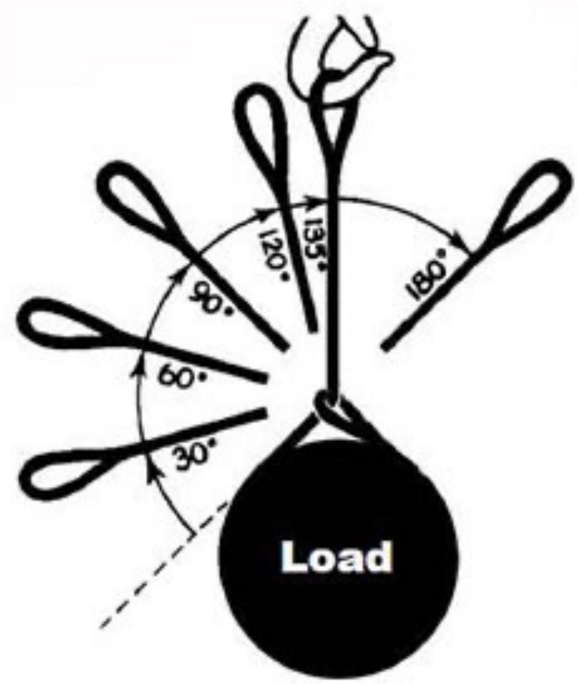
Choker Sling Applications
Rating Adjustment

Choker Hitch Rated Capacity Adjustment

If a load is hanging free, the normal choke angle is approximately 135 degrees. When the angle is less than 135 degrees, an adjustment in the sling-rated capacity must be made.

Choker hitches at angles greater than 135 degrees are not recommended since they are unstable. Extreme care should be taken to determine the angle of choke as accurately as possible.

In controlled tests, where the angle was less than 120 degrees, the sling body always failed at the point of choke when pulled to destruction. Allowance for this phenomenon must be made anytime a choker hitch is used to shift, turn or control a load, or when the pull is against the choke in a multi-leg lift.



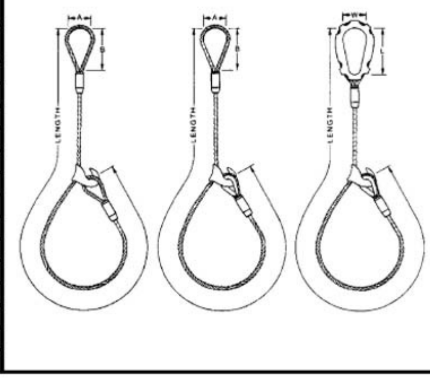
For wire rope slings in choker hitch when angle of choke is less than 120°.

Angle of Choke (Degrees)	Rated Capacity Percent *
Over - 120	100
90 - 120	87
60 - 89	74
30 - 59	62
0 - 29	49

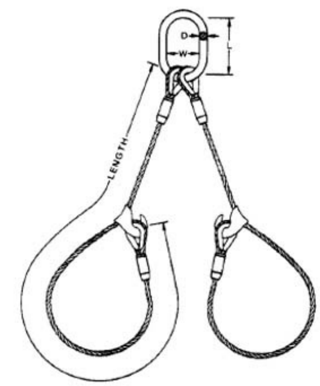
* Percent of sling rated capacity in a choker hitch.

Mechanically Spliced
Slings with Choker Hooks

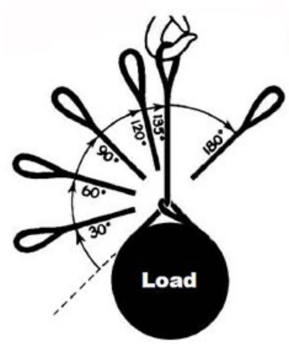
Rope Diameter 6 x 19 IWRC 6 x 36 IWRC	Min. Sling Length When Using Std. Eyes	Rated Capacity Tons *	Eye Dimensions		Slip-Thru Thimble		
					Number	Dimensions	
			A	B		L	W
1/4	1' 6"	.48	2	4	W-2	4-1/8	2-1/8
5/16	1' 9"	.74	2-1/2	5	W-2	4-1/8	2-1/8
3/8	2' 0"	1.1	3	6	W-2	4-1/8	2-1/8
7/16	2' 3"	1.4	3-1/2	7	W-3	4-3/8	2-3/8
1/2	2' 6"	1.9	4	8	W-3	4-3/8	2-3/8
9/16	2' 9"	2.4	4-1/2	9	W-3	4-3/8	2-3/8
5/8	3' 0"	2.9	5	10	W-4	6-5/8	3-3/8
3/4	3' 6"	4.1	6	12	W-4	6-5/8	3-3/8
7/8	4' 0"	5.6	7	14	W-5	7-1/8	3-3/4
1	4' 6"	7.2	8	16	W-5	7-1/8	3-3/4
1-1/8	5' 0"	9.1	9	18	W-6	8-3/8	4-3/8
1-1/4	5' 6"	11	10	20	W-6	8-3/8	4-3/8
1-3/8	6' 0"	13	11	22	W-7	9-1/2	5
1-1/2	7' 0"	16	12	24	W-7	9-1/2	5



Rope Diameter 6 x 19 IWRC 6 x 36 IWRC	Min. Sling Length When Using Std. Eyes	Rated Capacity—Tons * 2-Leg Choker Hitch ***			Alloy Oblong Link		
		60°	45°	30°	D	L	W
1/4	1' 6"	1.1	.91	.65	1/2	5	2-1/2
5/16	1' 9"	1.7	1.4	1	5/8	6	3
3/8	2' 0"	2.5	2	1.4	3/4	5-1/2	2-3/4
7/16	2' 3"	3.4	2.7	19.	7/8	6	3
1/2	2' 6"	4.4	3.6	2.5	3/4	5-1/2	2-3/4
9/16	2' 9"	5.5	4.5	3.2	1	8	4
5/8	3' 0"	6.8	5.5	3.9	1	8	4
3/4	3' 6"	9.7	7.9	5.6	1	8	4
7/8	4' 0"	13	11	7.6	1-1/4	8-3/4	4-3/8
1	4' 6"	17	14	9.8	1-1/4	8-3/4	4-3/8
1-1/8	5' 0"	21	17	12	1-1/2	12	6
1-1/4	5' 6"	26	21	15	1-3/4	12	6
1-3/8	6' 0"	31	25	18	1-3/4	12	6
1-1/2	7' 0"	37	30	21	2	14	7



- *Rated capacities in tons – 2,000 lbs per ton.
- ***See Choker Hitch rated capacity adjustment.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Sling angles less than 30° shall not be used.
- Rated capacity for 2-legged bridles, whether used as chokers or with hooks or other end fixtures, is affected by rigging angles, the same as straight slings in basket hitches.
- Note: Reduction in rated capacity as legs spread to wider lifting connections.



For wire rope slings in choker hitch when angle of choke is less than 120°.

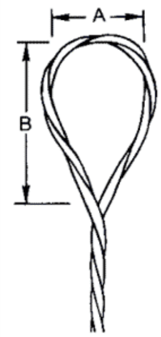
Angle of Choke (Degrees)	Rated Capacity Percent *
Over - 120	100
90 - 120	87
60 - 89	74
30 - 59	62
0 - 29	49

* Percent of sling rated capacity in a choker hitch.

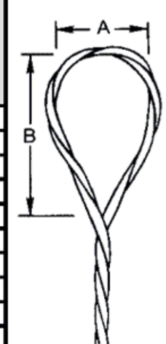
Braided

6-Part & 8-Part Slings

Rope Diameter (in.)	Min. Sling Length (Std. Eyes)	Width of Body (in.)	Thickness of Body (in.)	Vertical	Choker** Hitch	Basket Hitch**	Basket Hitch** 60°	Basket Hitch** 45°	Basket Hitch** 30°	Standard Eye Dimension (in.) A	Standard Eye Dimension (in.) B	Slip Thru Thimble ST (in.)	Heavy Thimble HT (in.)
Rated Capacity in Tons (2000 lbs.) *													
3/32	2' 0"	7/16	1/4	.42	.37	.84	.73	.59	.42	2	4	W-2	1/4
1/8	2' 0"	9/16	3/8	.84	.74	1.7	1.5	1.2	.84	3	6	W-2	5/16
3/16	3' 0"	13/16	1/2	1.6	1.4	3.2	2.8	2.3	1.6	4	8	W-3	1/2
1/4	3' 6"	1-1/8	11/16	2.9	2.5	5.7	4.9	4	2.9	5	10	W-4	5/8
5/16	4' 6"	1-3/8	7/8	4.4	3.9	8.9	7.7	6.3	4.4	6	12	W-4	3/4
3/8	5' 0"	1-11/16	1	6.3	5.5	13	11	9	6.3	7	14	W-5	7/8
7/16	6' 0"	2	1-3/16	8.6	7.5	17	15	12	8.6	8	16	W-5	1
1/2	6' 6"	2-1/4	1-5/16	11	9.8	22	19	16	11	9	18	W-6	1-1/8
9/16	7' 0"	2-1/2	1-1/2	14	12	28	24	20	14	10	20	W-6	1-3/8
5/8	8' 0"	2-13/16	1-11/16	17	15	35	30	24	17	11	22	W-7	1-1/2
3/4	9' 0"	3-3/8	2	25	22	49	43	35	25	12	24	W-8	1-5/8
7/8	10' 0"	4	2-5/16	33	29	67	58	47	33	14	28	W-9	2
1	11' 0"	4-1/2	2-11/16	43	38	87	75	61	43	16	32	W-10	–



Rope Diameter (in.)	Min. Sling Length (Std. Eyes)	Sling Diameter (in.)	Vertical	Choker** Hitch	Basket Hitch**	Basket Hitch** 60°	Basket Hitch** 45°	Basket Hitch** 30°	Standard Eye Dimension (in.) A	Standard Eye Dimension (in.) B	Slip Thru Thimble ST (in.)	Heavy Thimble HT (in.)
Rated Capacity in Tons (2000 lbs.) *												
3/32	2' 0"	7/16	.56	.49	1.1	.97	.79	.56	2	4	W-2	5/16
1/8	2' 0"	9/16	1.1	1	2.2	1.9	1.6	1.1	3	6	W-2	3/8
3/16	3' 0"	13/16	2.2	1.9	4.3	3.7	3	2.2	4	8	W-3	1/2
1/4	3' 6"	1-1/8	3.8	3.3	7.6	6.6	5.4	3.8	5	10	W-4	3/4
5/16	4' 6"	1-3/8	5.9	5.2	12	10	8.3	5.9	6	12	W-5	1
3/8	5' 0"	1-11/16	8.5	7.4	17	15	12	8.5	7	14	W-5	1-1/8
7/16	6' 0"	2	11	10	23	20	16	11	8	16	W-6	1-1/4
1/2	6' 6"	2-1/4	15	13	30	26	21	15	9	18	W-7	1-3/8
9/16	7' 0"	2-1/2	19	16	38	33	27	19	10	20	W-7	1-1/2
5/8	8' 0"	2-13/16	23	20	46	40	33	23	11	22	W-8	1-3/4
3/4	9' 0"	3-3/8	33	29	66	57	47	33	12	24	W-9	2
7/8	10' 0"	4	45	39	89	77	63	45	14	28	W-10	–
1	11' 0"	4-1/2	58	51	116	100	82	58	16	32	W-10	–

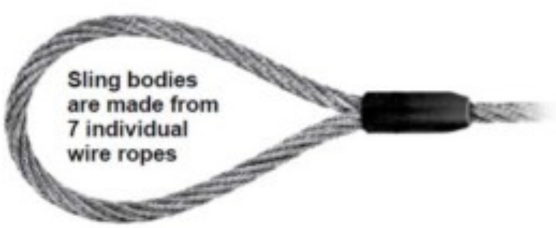


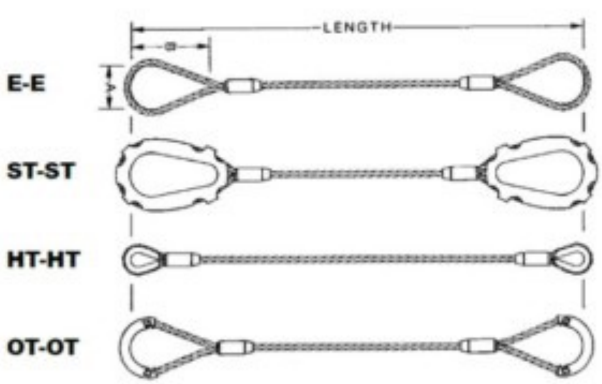
- Sling capacities for rope diameters 1/8" thru 5/16" based on 7x19 galvanized cable.
- *Rated capacities in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 15.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Sling angles less than 30° shall not be used.
- **See Choker Hitch rated capacity adjustment.

Mechanically Spliced
Cable Laid Slings

Rated Capacity in Tons (2000 lbs.) *												
Rope Diameter (in.)	Min. Sling Length (Std. Eyes)	Vertical Capacity	Choker ** Capacity	Basket Hitch**				Standard Eye Dimension		Slip Thru Thimble ST	Heavy Thimble HT (in.)	Slip-On Thimble QT (in.)
					60°	45°	30°	A	B			
1/4	1' 6"	0.5	.35	1	.87	.71	.50	2	4	W-2	1/4	3/8
3/8	2' 0"	1.1	.8	2.2	1.9	1.5	1.1	3	6	W-2	3/8	3/8
1/2	2' 6"	1.9	1.3	3.7	3.2	2.6	1.9	4	8	W-3	1/2	1/2
5/8	3' 0"	2.8	1.9	5.5	4.8	3.9	2.8	5	10	W-4	5/8	5/8
3/4	3' 6"	3.8	2.7	7.6	6.6	5.4	3.8	6	12	W-4	3/4	3/4
7/8	4' 0"	5	3.5	10	8.7	7.1	5	7	14	W-5	7/8	7/8
1	4' 6"	6.4	4.5	13	11	9.1	6.4	8	16	W-5	1	1
1-1/8	5' 0"	8.3	5.8	17	14	12	8.3	9	18	W-6	1-1/8	–
1-1/4	5' 6"	9.9	6.9	20	17	14	9.9	10	20	W-6	1-1/4	–

- *Rated capacities in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 10.
- Rated capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated capacities based on design factor of 5.
- Sling angles less than 30° shall not be used.
- **See Choker Hitch rated capacity adjustment.

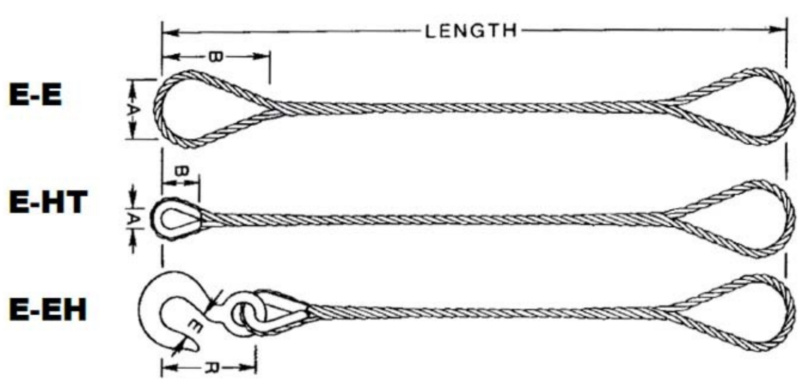


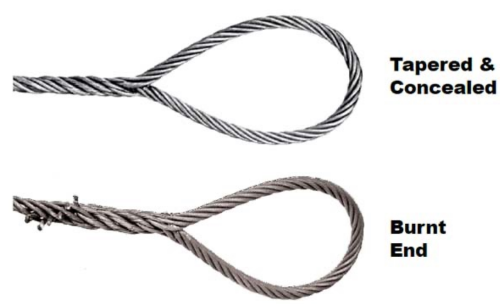


Hand Spliced
1-Leg Slings

Construction	Rope Diameter (inches)	Min. Sling Length When Using Std. Eyes	Rated Capacity—Tons *						Eye Dimensions		Thimble		Hook		
			Vertical	*** Choker Hitch	Basket Hitch										
					U	60	45	30	A	B	A	B	** WLL	E	R
6 x 19 FC	1/4	2' 0"	.54	.42	1.1	.94	.77	.42	2	4	7/8	1-5/8	3/4	15/16	3-7/32
6 x 19 FC	5/16	2' 3"	.83	.66	1.7	1.4	1.2	.66	2-1/2	5	1-1/16	1-7/8	1	1-3/32	3-21/32
6 x 19 FC	3/8	2' 6"	1.2	.94	2.4	2	1.7	.94	3	6	1-1/8	2-1/8	1-1/2	1-1/16	4-3/32
6 x 19 FC	7/16	2' 9"	1.6	1.3	3.2	2.7	2.2	1.3	3-1/2	7	1-1/4	2-1/4	1-1/2	1-1/16	4-3/32
6 x 19 FC	1/2	3' 0"	2	1.6	4	3.5	2.9	1.6	4	8	1-1/2	2-3/4	2	1-7/32	4-11/16
6 x 19 FC	9/16	3' 6"	2.5	2.1	5	4.4	3.6	2.1	4-1/2	9	1-1/2	2-3/4	3	1-1/2	5-3/4
6 x 19 FC	5/8	4' 0"	3.1	2.6	6.2	5.3	4.4	2.6	5	10	1-3/4	3-1/4	3	1-1/2	5-3/4
6 x 19 FC	3/4	4' 6"	4.3	3.7	8.6	7.4	6.1	3.7	6	12	2	3-3/4	5	1-7/8	7-3/8
6 x 36 FC	7/8	5' 6"	5.7	5	11	9.8	8	5	6-1/2	13	2-1/4	4-1/4	7-1/2	2-1/4	9-1/16
6 x 36 FC	1	6' 0"	7.4	6.4	15	13	10	6.4	7	14	2-1/2	4-1/2	7-1/2	2-1/4	9-1/16
6 x 36 FC	1-1/8	6" 6"	9.3	8.1	19	16	13	8.1	7-1/2	15	2-7/8	5-1/8	10	2-1/2	10-1/16
6 x 36 FC	1-1/4	7' 9"	11	9.9	23	20	16	9.9	8	16	2-7/8	5-1/8	10	2-1/2	10-1/16
6 x 36 FC	1-3/8	9' 0"	14	12	27	24	19	12	8-1/2	17	3-1/2	6-1/4	15	3-3/8	12-1/2
6 x 36 FC	1-1/2	10' 0"	16	14	32	28	23	14	9	18	3-1/2	6-1/4	15	3-3/8	12-1/2

- *Rated capacities in tons – 2,000 lbs per ton.
- Basket Hitch based on D/d ratio of 15.
- Rated Capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated Capacities based on design factor of 5.
- Sling angles less than 30° shall not be used.
- **Working Load Limit in tons – 2,000 lbs per ton.
- ***See choker hitch rated capacity adjustment.
- **WARNING:** Hand-spliced wire rope slings should not be used in lifts where the sling may rotate and cause the wire rope to unlay.





Hand Tucked Spliced
Endless Grommet

Rated Capacity in Tons (2000 lbs.)						
Finished Sling Body Diameter (in.)	Vertical	Basket Hitch				
		Choker	Vertical	30°	45°	60°
1/4	0.94	0.66	1.9	1.6	1.3	0.94
5/16	1.5	1	2.9	2.5	2.1	1.5
3/8	2.1	1.5	4.2	3.6	3	2.1
7/16	2.8	2	5.7	4.9	4	2.8
1/2	3.7	2.6	7.3	6.4	5.2	3.7
9/16	4.6	3.2	9.3	8	6.6	4.6
5/8	5.7	4	11	9.9	8.1	5.7
3/4	8.2	5.7	16	14	12	8.2
7/8	11	7.7	22	19	16	11
1	14	10	29	25	20	14
1-1/8	18	12	35	31	25	18
1-1/4	21	15	43	37	30	21
1-3/8	25	18	51	44	36	25
1-1/2	30	21	60	52	42	30
1-5/8	34	24	69	60	49	34
1-3/4	40	28	79	69	56	40
1-7/8	45	31	89	77	63	45
2	50	35	101	87	71	50
2-1/8	56	39	112	97	79	56
2-1/4	62	43	124	107	88	62
2-3/8	68	48	137	118	97	68
2-1/2	75	52	149	129	106	75
2-3/4	89	62	177	154	125	89
3	104	73	207	180	147	104

- Rated capacities in tons – 2,000 lbs per ton.
- Rated capacities using 6x19 and 6x36 Class EIPS fiber core rope.
- Basket Hitch and Vertical Lift based on D/d ratio of 5 where "d" = body diameter of the finished grommet and "D" is the diameter of the pin..
- Rated Capacities based on pin diameter no smaller than 5 times the body diameter.
- Rated Capacities based on design factor of 5.
- Horizontal sling angles less than 30° shall not be used.

Mechanically Spliced
Endless Grommet

Rated Capacity in Tons (2000 lbs.)						
Finished Sling Body Diameter (in.)	Vertical	Basket Hitch				
		Choker	Vertical	30°	45°	60°
1/4	1.1	.74	2.1	1.1	1.5	1.8
5/16	1.6	1.2	3.3	1.6	2.3	2.8
3/8	2.4	1.6	4.7	2.4	3.3	4.1
7/16	3.2	2.2	6.4	3.2	4.5	5.5
1/2	4.1	2.9	8.3	4.1	5.9	7.2
9/16	5.2	3.7	10	5.2	7.4	9.1
5/8	6.4	4.5	13	6.4	9.1	11
3/4	9.2	6.4	18	9.2	13	16
7/8	12	8.7	25	12	18	22
1	16	11	32	16	23	28
1-1/8	20	14	41	20	29	35
1-1/4	25	17	50	25	35	43
1-3/8	30	21	60	30	42	52
1-1/2	36	25	71	36	50	62
1-5/8	41	29	82	41	58	71
1-3/4	48	33	95	48	68	83
1-7/8	54	38	109	54	77	94
2	62	43	124	62	87	107
2-1/8	69	48	138	69	98	119
2-1/4	77	54	154	77	109	133
2-3/8	85	60	171	85	121	148
2-1/2	94	66	188	94	133	163
2-3/4	113	79	225	113	159	195
3	133	93	265	133	188	230

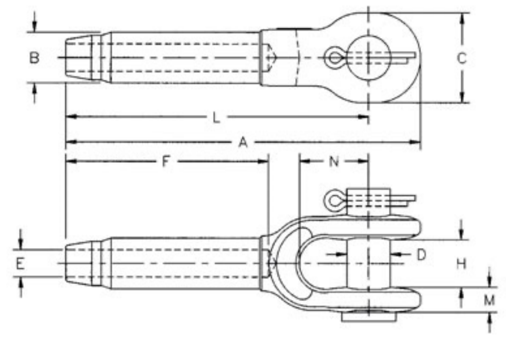
- Rated capacities in tons – 2,000 lbs per ton.
- Rated capacities using 6x19 and 6x36 Class EIPS IWRC wire rope.
- Basket Hitch and Vertical Lift based on D/d ratio of 5 where “d” = body diameter of the finished grommet and “D” is the diameter of the pin..
- Rated Capacities based on pin diameter no smaller than 5 times the body diameter.
- Rated Capacities based on design factor of 5.
- Horizontal sling angles less than 30° shall not be used.

Swage Socket Assemblies

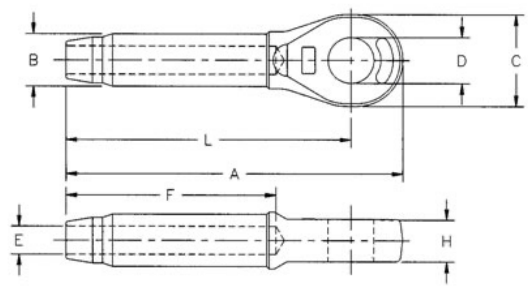
Open & Closed Sockets

Boom Pendant Lines

Wire Rope Diameter (in.)	Open Swage Socket Dimensions & Data (in.)									
	A	B	C	D	E	F	H	L	M	N
1/4	4.81	.50	1.38	.69	.27	2.13	.69	4.00	.38	1.50
5/16	6.25	.77	1.62	.81	.34	3.19	.81	5.31	.47	1.75
3/8	6.25	.77	1.62	.81	.41	3.19	.81	5.31	.47	1.75
7/16	7.81	.98	2.00	1.00	.48	4.25	1.00	6.69	.56	2.00
1/2	7.81	.98	2.00	1.00	.55	4.25	1.00	6.69	.56	2.00
9/16	9.50	1.25	2.38	1.19	.61	5.31	1.25	8.13	.68	2.25
5/8	9.50	1.25	2.38	1.19	.67	5.31	1.25	8.13	.68	2.25
3/4	11.56	1.55	2.75	1.38	.80	6.38	1.50	10.00	.78	2.75
7/8	13.41	1.70	3.13	1.62	.94	7.44	1.75	11.63	.94	3.25
1	15.47	1.98	3.69	2.00	1.06	8.50	2.00	13.38	1.06	3.75
1-1/8	17.31	2.25	4.06	2.25	1.19	9.56	2.25	15.00	1.19	4.25
1-1/4	19.06	2.53	4.50	2.50	1.33	10.63	2.50	16.50	1.22	4.75
1-3/8	20.94	2.80	5.00	2.50	1.45	11.69	2.50	18.13	1.38	5.25
1-1/2	22.88	3.08	5.50	2.75	1.58	12.75	3.00	19.75	1.69	5.75
1-3/4	26.63	3.39	6.69	3.50	1.86	14.88	3.50	23.00	2.11	6.75
2	31.44	3.94	8.00	3.75	2.11	17.00	4.00	26.88	2.37	8.00

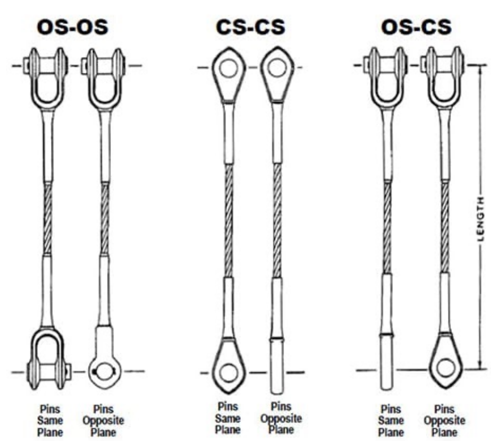


Wire Rope Diameter (in.)	Closed Swage Socket Dimensions & Data (in.)							
	A	B	C	D	E	F	H	L
1/4	4.31	.50	1.38.	.75	.27	2.12	.50	3.50
5/16	5.44	.77	1.62	.88	.34	3.19	.67	4.50
3/8	5.44	.77	1.62	.88	.41	3.19	.67	4.50
7/16	6.91	.98	2.00	1.06	.48	4.25	.86	5.75
1/2	6.91	.98	2.00	1.06	.55	4.25	.86	5.75
9/16	8.66	1.25	2.38	1.25	.61	5.31	1.13	7.25
5/8	8.66	1.25	2.38	1.25	.67	5.31	1.13	7.25
3/4	10.28	1.55	2.88	1.44	.80	6.38	1.31	8.63
7/8	11.94	1.70	3.12	1.69	.94	7.44	1.50	10.13
1	13.56	1.98	3.63	2.06	1.06	8.50	1.75	11.50
1-1/8	15.03	2.25	4.00	2.31	1.19	9.56	2.00	12.75
1-1/4	16.94	2.53	4.50	2.56	1.33	10.63	2.25	14.38
1-3/8	18.63	2.80	5.00	2.56	1.45	11.69	2.25	15.75
1-1/2	20.12	3.08	5.50	2.81	1.58	12.75	2.50	17.00
1-3/4	23.56	3.69	6.25	3.56	1.86	14.88	3.00	20.00
2	27.62	3.94	7.25	3.81	2.11	17.00	3.25	23.00



Rated Capacity—Tons		
Wire Rope Diameter (in.)	6 x 19 & 6 x 36 IWRC	
	EIP	EEIP
1/4	.68	.74
5/16	1.1	1.2
3/8	1.5	1.7
7/16	2	2.2
1/2	2.7	2.9
9/16	3.4	3.7
5/8	4.1	4.5
3/4	5.9	6.5
7/8	8	8.8
1	10	11
1-1/8	13	14
1-1/4	16	18
1-3/8	19	21
1-1/2	23	25
1-3/4	31	34
2	40	43

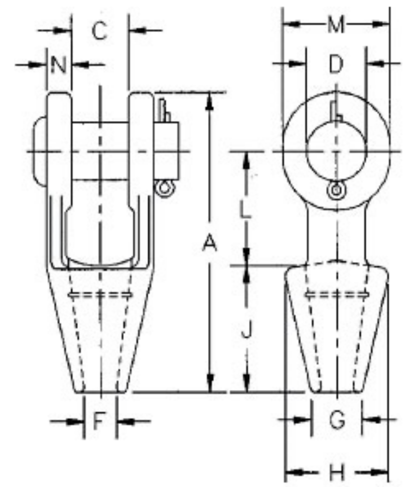
Swage socket assemblies have a 100% termination efficiency meaning the ultimate strength of the assembly equals the breaking strength of the wire rope based on IWRC EIPS rope.



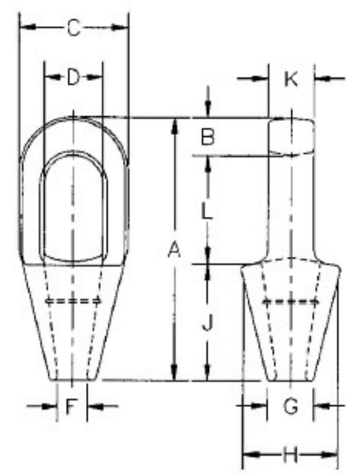
Spelter Socket Assemblies

Open & Closed Sockets
Boom Pendant Lines

Wire Rope Diameter (in.)	Open Spelter Socket Dimensions & Data (in.)									
	A	C	D	F	G	H	J	L	M	N
1/4	4.56	.91	.69	.38	.69	1.56	2.25	1.56	1.31	.36
5/16 to 3/8	4.84	.81	.81	.50	.81	1.69	2.25	1.75	1.50	.44
7/16 to 1/2	5.56	1	1	.56	.94	1.88	2.50	2.00	1.88	.50
9/16 to 5/8	6.75	1.25	1.19	.69	1.13	2.25	3	2.50	2.25	.56
3/4	7.94	1.50	1.38	.81	1.25	2.62	3.50	3	2.62	.62
7/8	9.25	1.75	1.63	.94	1.50	3.25	4	3.50	3.13	.80
1	10.56	2	2	1.13	1.75	3.75	4.50	4	3.75	.88
1-1/8	11.81	2.25	2.25	1.25	2	4.12	5	4.62	4.12	1
1-1/4 to 1-3/8	13.19	2.50	2.50	1.50	2.25	4.75	5.50	5	4.75	1.13
1-1/2	15.12	3	2.75	1.63	2.75	5.25	6	6	5.38	1.19
1-5/8	16.25	3	3	1.75	3	5.50	6.50	6.50	5.75	1.31
1-3/4 to 1-7/8	18.25	3.50	3.50	2	3.13	6.38	7.50	7	6.50	1.56
2 to 2-1/8	21.50	4	3.75	2.25	3.75	7.38	8.50	9	7.00	1.81
2-1/4 to 2-3/8	23.50	4.50	4.25	2.50	4	8.25	9	10	7.75	2.13
2-1/2 to 2-5/8	25.50	5	4.75	2.88	4.50	9.25	9.75	10.75	8.50	2.38
2-3/4 to 2-7/8	27.25	5.25	5	3.12	4.88	10.50	11	11	9	2.88
3 to 3-1/8	29	5.75	5.25	3.38	5.25	11.12	12	11.25	9.50	3.00
3-1/4 to 3-3/8	30.88	6.25	5.50	3.62	5.75	11.88	13	11.75	10	3.12
3-1/2 to 3-5/8	33.25	6.75	6	3.88	6.50	12.38	14	12.50	10.75	3.25
3-3/4 to 4	36.25	7.50	7	4.25	7.25	13.62	15	13.50	12.50	3.50

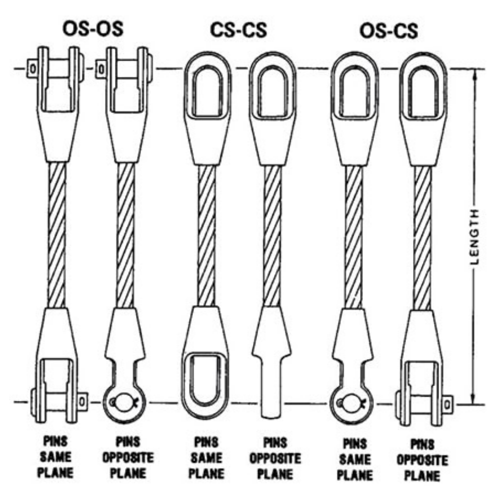


Wire Rope Diameter (in.)	Closed Spelter Socket Dimensions & Data (in.)									
	A	B	C	D	F	G	H	J	K	L
1/4	4.50	.50	1.50	.88	.38	.69	1.56	2.25	.50	1.75
5/16 to 3/8	4.88	.62	1.69	.97	.50	.81	1.69	2.25	.69	2
7/16 to 1/2	5.44	.69	2	1.16	.56	.94	1.88	2.50	.88	2.25
9/16 to 5/8	6.31	.81	2.63	1.41	.69	1.12	2.38	3	1.00	2.50
3/4	7.56	1.06	3	1.66	.81	1.25	2.75	3.56	1.25	3
7/8	8.75	1.25	3.63	1.88	.94	1.50	3.25	4	1.50	3.50
1	9.88	1.38	4.13	2.30	1.13	1.75	3.75	4.44	1.75	4
1-1/8	11	1.50	4.50	2.56	1.25	2.00	4.13	5	2.00	4.50
1-1/4 to 1-3/8	12.12	1.63	5.30	2.81	1.50	2.25	4.75	5.50	2.25	5
1-1/2	13.94	1.94	5.33	3.19	1.63	2.75	5.25	6	2.50	6
1-5/8	15.13	2.13	5.75	3.25	1.75	3.00	5.50	6.50	2.75	6.50
1-3/4 to 1-7/8	17.25	2.19	6.75	3.75	2	3.13	6.38	7.50	3	7.56
2 to 2-1/8	19.50	2.44	7.63	4.38	2.25	3.75	7.38	8.50	3.25	8.56
2-1/4 to 2-3/8	21.13	2.63	8.50	5	2.50	4	8.25	9	3.63	9.50
2-1/2 to 2-5/8	23.50	3.12	9.50	5.50	2.88	4.50	9.25	9.75	4	10.62
2-3/4 to 2-7/8	25.38	3.12	10.75	6.25	3.12	4.88	10.19	11	4.88	11.25
3 to 3-1/8	27	3.25	11.50	6.75	3.38	5.25	11.50	12	5.25	11.75
3-1/4 to 3-3/8	29.25	4	12.25	7.25	3.62	5.75	12.25	13	5.75	12.25
3-1/2 to 3-5/8	31	4	13	7.75	3.88	6.50	13	14	6.25	13
3-3/4 to 4	33.25	4.25	14.25	8.50	4.25	7.25	14.25	15	7	14



Rated Capacity—Tons		
Wire Rope Diameter (in.)	6 x 19 & 6 x 36 IWRC	
	EIP	EEIP
1/2	2.7	2.9
9/16	3.4	3.7
5/8	4.1	4.5
3/4	5.9	6.5
7/8	8	8.8
1	10	11
1-1/8	13	14
1-1/4	16	18
1-3/8	19	21
1-1/2	23	25
1-5/8	26	29
1-3/4	31	34
1-7/8	35	38
2	40	43
2-1/8	44	49
2-1/4	49	54
2-3/8	55	60
2-1/2	60	66
2-5/8	66	73
2-3/4	72	79
2-7/8	78	86
3	85	94
3-1/8	92	101
3-1/4	98	109
3-3/8	106	116
3-1/2	113	124

Spelter socket assemblies have a 100% termination efficiency meaning the ultimate strength of the assembly equals the breaking strength of the wire rope based on IWRC EIPS rope.



Sling Inspections Guidelines

As a starting point, the same work practices which apply to all "working" wire rope apply to wire rope which has been fabricated into a sling. Therefore, a good working knowledge of wire rope design and construction will not only be useful, but essential in conducting a wire rope sling inspection.

There are two industry standards that exist to provide the end-user with guidelines for inspection and criteria that warrants removal from service: *OSHA 1910.184* and *ASME B30.9*.

Initial Inspection (prior to initial use): Best practice is to inspect the wire rope sling upon receiving it from the manufacturer. Double-check the sling tag to make sure it's what you ordered and that the rated capacity meets all of your project specifications and lifting requirements.

Frequent (daily or prior to use): Designate a Competent Person to perform a daily visual inspection of slings and all fastenings and attachments for damage, defects, or deformities. The inspector should also make sure that the wire rope sling that was selected meets the specific job requirements it's being used for.

Users can't rely on a once-a-day inspection if the wire rope sling is used multiple times throughout the day. Damage to wire rope can occur on one lift and best practice is to perform a visual inspection before any shift change or changes in lifting application. Because shock loads, severe angles, sharp edges, and excessive heat can quickly cause damage to a lifting sling, the user should inspect the sling prior to each lift.

Sling Inspections
Guidelines - Continued

Periodic Inspection: A periodic inspection is performed by either a professional service provider, or by a Qualified Person, every 12 months (at a minimum) and monthly to quarterly in more severe service conditions. The following are all determining factors in scheduling the frequency of a periodic inspection:

- Frequency of use
- Severity of service conditions
- Nature of the lifts being performed
- Experience gained on the service life of wire rope slings used in similar applications
- ASME provides these additional periodic inspection guidelines based on the service of the wire rope sling:
- Normal Service – Yearly
- Severe Service – Monthly to Quarterly
- Special Service – As recommended by a Qualified Person

Depending on the severity of the operating environment and frequency of use, your business may decide to inspect wire rope slings more often than the minimum yearly requirement.

Periodic inspections are required to be documented per *ASME B30.9* and records of those inspections retained.

The employer is required to maintain a record of the most recent thorough inspection—however, individual records for each sling that was inspected are not required. Failure to maintain and retain inspection records is one of the most common issues we see that can prevent a company from reaching full OSHA compliance.